

Implementing Traffic Signal Optimal Control by Multiagent Reinforcement Learning

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Abstract—In urban traffic environment, the traffic flow is more difficult to be predicted properly because the interaction and intertwinement among multiple crossroads, which makes preset traffic control model can not keep always high performance in all traffic situations. Considering the capability of autonomous learning being inherent in reinforcement learning, we propose a multiagent reinforcement learning based traffic signal control method. Without preset control model, multiple collaborative agents can learn the optimal control policy corresponding real traffic situation. The experiment results demonstrate the applicability and effectiveness of our approach.

Keywords- traffic signal control; multi-agent reinforcement learning; autonomous learning; multi-crossroads urban traffic; traffic flow

I. INTRODUCTION

To implement intelligent traffic signal control is one of the low cost ways for reducing traffic congestion and improving traffic efficiency [1]. But to model the traffic flow is still very difficult, because it is time-varying and stochastic, especially in multi-crossroads urban traffic environment. A preset traffic control model is impossible to fulfill all traffic situations. So more attentions are paid to machine learning on real time traffic flow control [2] [3] [4]. In all these methods of machine learning, reinforcement learning gets more focus, because the capability of autonomous learning, which makes agent can perceive the varying condition and self-adaptively adjust the control policy in order to correspond to traffic conditions.

A Q-learning based adaptive traffic signal control for minimizing delay time is proposed in [2], but only one crossroad is considered. Salkham [3] presented a Reinforcement Learning framework for traffic control optimization, but existing more dependency on hardware infrastructure. Li[4] proposed an adaptive dynamic programming schema to implement multi-crossroads traffic signal intelligent control, but the interaction and intertwined among multiple crossroads are not discussed in detail. Some similar works can be found in [5] [6] [7]. These works express to use the autonomous learning property of reinforcement learning to solve the time-varying traffic flow problem has been one of research focus in intelligent traffic signal control[8]. But some problems keep open and are not discussed in detail.

In this work, we present a multiagent reinforcement learning based multi-crossroads traffic signal control model. First, we define the multiagent reinforcement learning model for the multi-crossroads traffic signal control problem, which include state space, goal state, action, and training episode, to give complete problem description in reinforcement learning language. Then, the multiagent reinforcement learning algorithm to discover the optimal policy for multi-crossroad traffic signal control is presented. The optimal traffic signal control policy is the policy to make all vehicles can leave the urban traffic network which constructed by multiple crossroads in the shortest time. At last, in the simulation environment, we compare the time of all vehicles to leave the urban traffic network between conventional fixed-time control and multiagent reinforcement learning based adaptive control to demonstrate the affectivity of our approach.

II. THE REINFORCEMENT LEARNING MODEL FOR THE MULTI-CROSSROADS TRAFFIC SIGNAL CONTROL PROBLEM

First we give the multiagent reinforcement learning model description for the multi-crossroad traffic signal control problem.

In our model, an urban traffic network is constructed by multiple crossroads, for example, Figure.1 shows a traffic network constructed by 6 crossroads, and it is also the simulation environment we implemented. Of course, the traffic network can be enlarged by more crossroads.

The goal of traffic signal control is make these vehicles on the crossroads to leave traffic network safely and quickly. It is obvious that the vehicles on the crossroads are time-varying and stochastic. The preset control model is impossible to adapt all traffic flows, and this is also the reason we take multiagent reinforcement learning method, which can learn autonomously the optimal policy according to real-time traffic flow.

In traffic signal control, a phase is time interval that a signal lamp are lighting continuously, for example, a lighting red lamp time interval or a lighting green lamp time interval both are phase. By switching the different phase in a reasonable time interval to make vehicles cross the crossroad in shorter time is the major goal of traffic signal control. For example, when the vehicles in east west direction is more than the vehicles in south north direction, then the green lamp phase on the east west direction should be long than the red lamp phase.

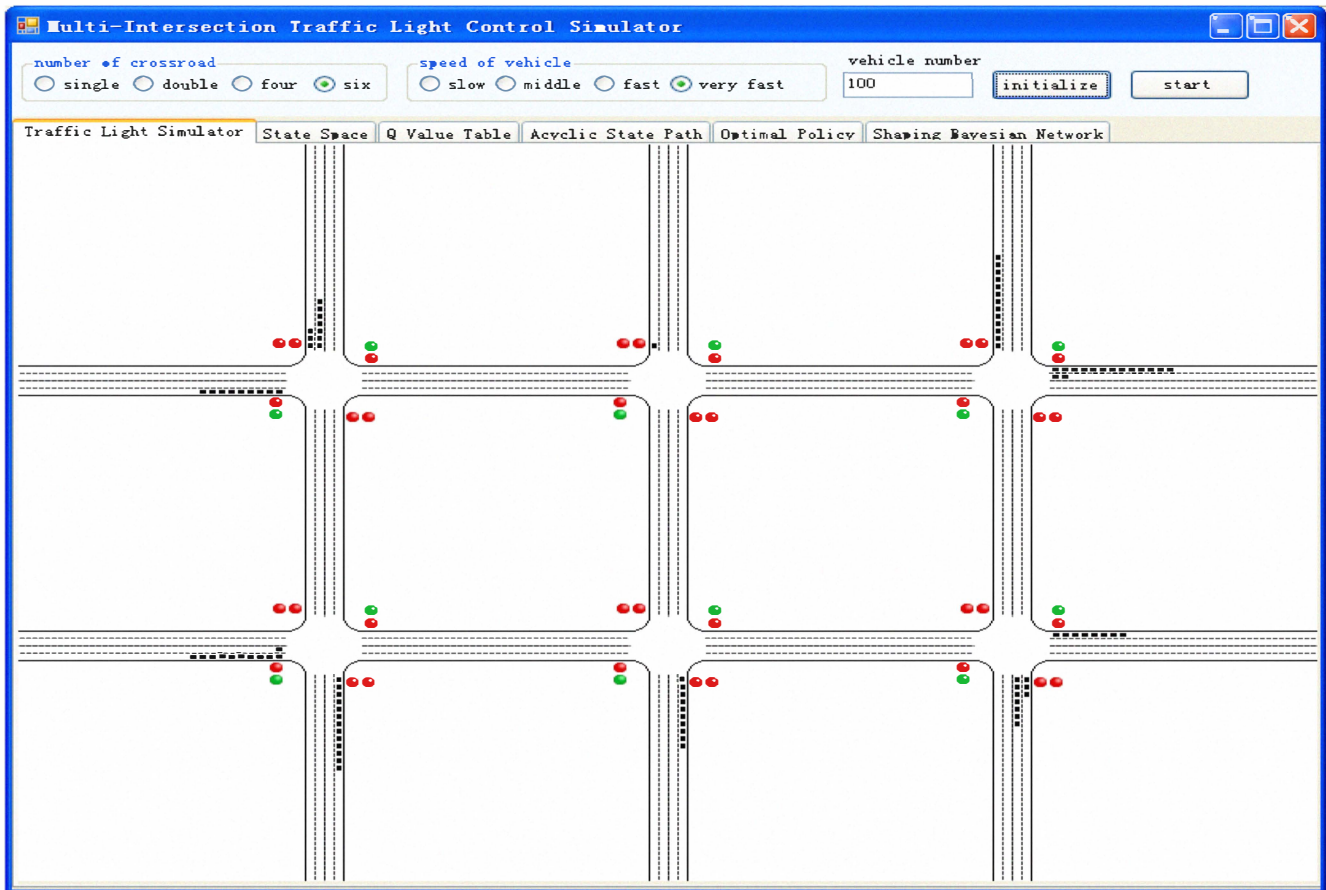


Figure 1. The traffic network constructed by 6 cross-roads in the urban traffic simulation environment

Usually the phase time interval is the integer times of a shortest phase time interval, noted as “MinPT”. For example, if MinPT=30 second, then the phase time interval should be 30 second, 60 second, 90 second, or 120 second, and so on.

According to the description of traffic signal control problem, the vehicle numbers on all crossroads are combined to be a state, and represent a traffic flow shot. The control goal is to make all vehicles leaving the urban traffic network in shorter time, so when the numbers of every crossroad in the traffic network equal to zero, then the goal state is reached. After finish each shortest phase time interval, control agent can select to light red lamp or green lamp. When one direction is set to light green lamp, then another direction must be set to light red lamp, because only one direction's vehicles can pass in one time. So the action after each shortest phase time interval is just to set one direction's green lamp, because another direction must be red lamp.

The whole training episode can be described as: After finish each shortest phase time interval, control agent select randomly one direction to light green lamp(another direction to light red lamp) to permit vehicles pass or not, and reach another state, then select next action, until all vehicles leave the traffic network(reach the goal state). This process is called a training episode. Fig.2 shows the state and action in a traffic network of two crossroads.

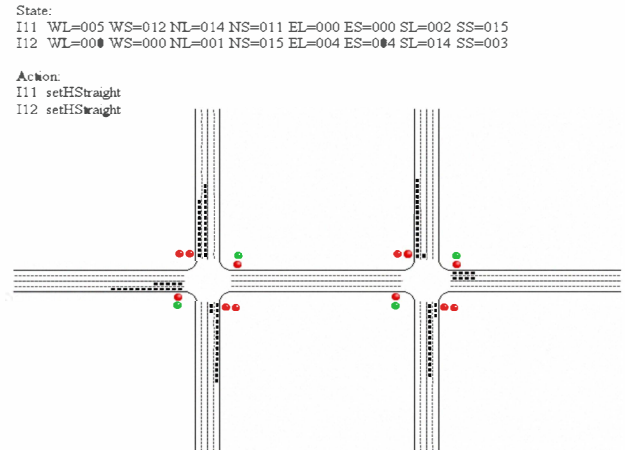


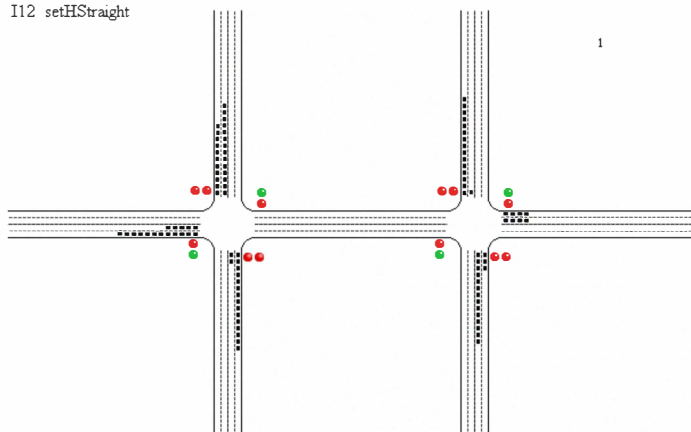
Figure 2. The state and action of traffic control problem

III. THE MULTIAGENT REINFORCEMENT LEARNING ALGORITHM TO FIND THE OPTIMAL POLICY FOR MULTI-CROSSROADS TRAFFIC SIGNAL CONTROL

We use the most widely used reinforcement learning method: Q-learning [9] to implement the algorithm to find the optimal policy for multi-crossroads traffic signal control. Algorithm 1 gives the formal procedure to find the optimal policy.

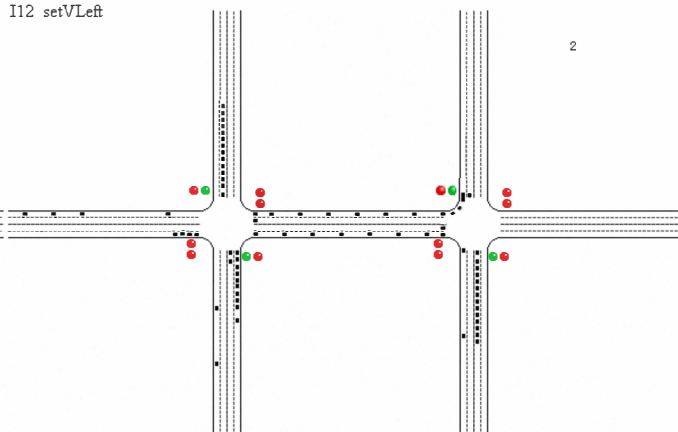
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I12 WL=000 WS=000 NL=001 NS=015 EL=004 ES=004 SL=014 SS=003

Action:
I11 setHSTraight
I12 setHSTraight



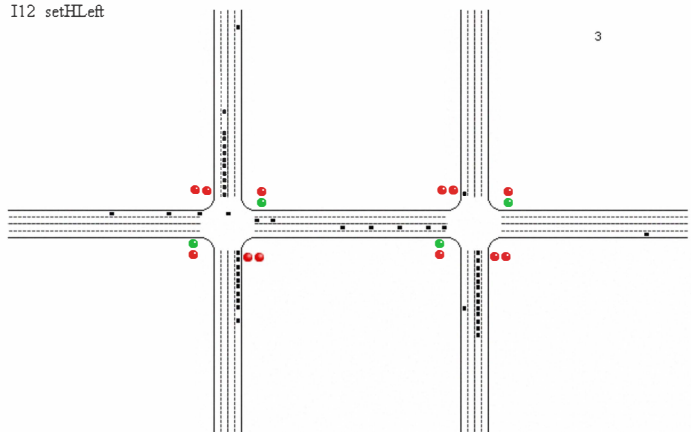
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Action:
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I12 setVLeft



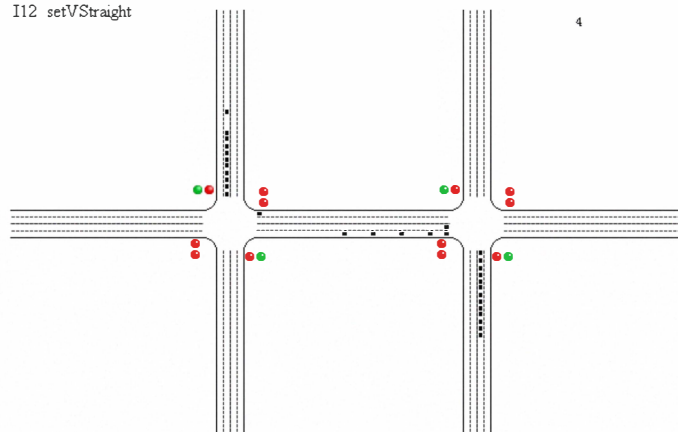
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Action:
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I12 setHLeft



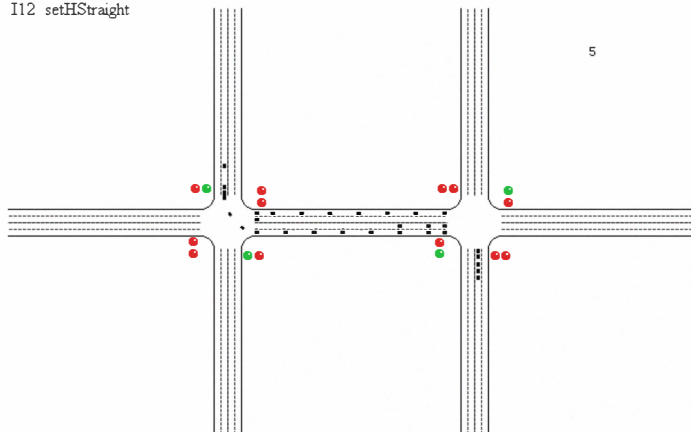
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Action:
I11 setVSTraight
I12 setVSTraight



State:
I11 WL=000 WS=000 NL=003 NS=000 EL=001 ES=008 SL=000 SS=000
I12 WL=003 WS=008 NL=000 NS=000 EL=000 ES=000 SL=005 SS=000

Action:
I11 setVLeft
I12 setHSTraight



State:
I11 WL=000 WS=000 NL=000 NS=000 EL=000 ES=000 SL=000 SS=000
I12 WL=000 WS=000 NL=000 NS=000 EL=000 ES=000 SL=000 SS=000

Action:
I11 setHLeft
I12 setHLeft

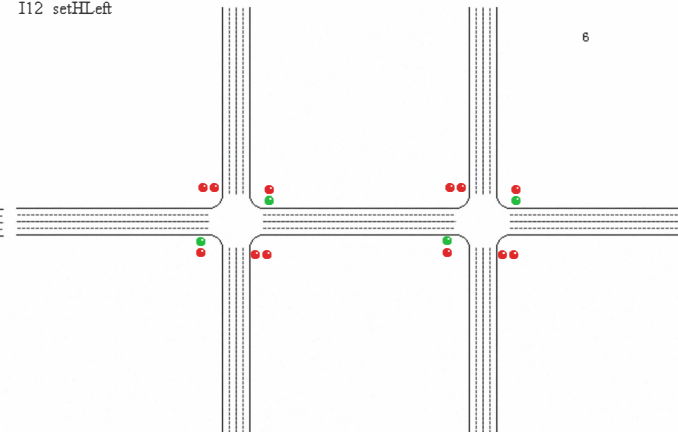


Figure.3 The optimal policy in 2 crossroads traffic network

Algorithm.1 Finding the optimal policy for traffic signal control problem by multiagent reinforcement learning

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Create null list: trajectory
Create null list: State-List
Create null set: State-Cluster

For each  $s, a$  initialize the table entry  $\hat{Q}(s, a)$  to zero
Observe the current state  $s$ 
Add  $s$  into trajectory
Repeat
    Select an action  $a$  and execute it
    Receive immediate reward  $r$ 
    Observe the current state  $s'$ 
    Add  $s'$  into trajectory
    Let State-List  $\leftarrow$  the reverse of trajectory

    Update the table entry for  $\hat{Q}(s, a)$  as follows:
        
$$\hat{Q}(s, a) \leftarrow r + \gamma \max_a \hat{Q}(s', a')$$


    For each list cur_list in State-Cluster do
        Let  $i \leftarrow 2$ 
        While  $i < \text{the length of cur\_list}$ 
            Let  $s^{pre} \leftarrow \text{cur\_list}[i-1]$ 
            Let  $s^{now} \leftarrow \text{cur\_list}[i]$ 
            Let  $a^*$   $\leftarrow$  the action agent takes when it move
            from  $s^{now}$  to  $s^{pre}$ 
            
$$\hat{Q}(s^{now}, a^*) \leftarrow r + \gamma \max_{a'} \hat{Q}(s^{pre}, a')$$

            Let  $i \leftarrow i+1$ 
        End while
    End for
     $s \leftarrow s'$ 

Until all table entry  $\hat{Q}(s, a)$  are convergent

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Fig.3 gives the optimal policy in the traffic network of two crossroads. From fig.3, it is obvious that the green lamp time in south north direction is longer than the green lamp time in east west direction, because the vehicle numbers in south north direction is greater than the vehicle numbers in east west direction. By algorithm 1, the control agent can adapt its control policy to correspond the real time traffic situation to implement self-adaptive control.

IV. EXPERIMENT

In urban traffic simulation environment, we compare the time of all vehicles to leave the urban traffic network between conventional fixed-time control and multiagent reinforcement learning based adaptive control in 30 times different traffic situation. The result is shown in Fig.4. From the compare diagram, the time of all vehicles to leave the traffic network

can be reduced by 20-30% when adopt the optimal policy found by the multiagent reinforcement learning based traffic signal control method, when compared with the conventional fixed-time traffic signal control method. The result shows our method can significantly reduce the time of all vehicles to leave the urban traffic network even in such a small traffic network.

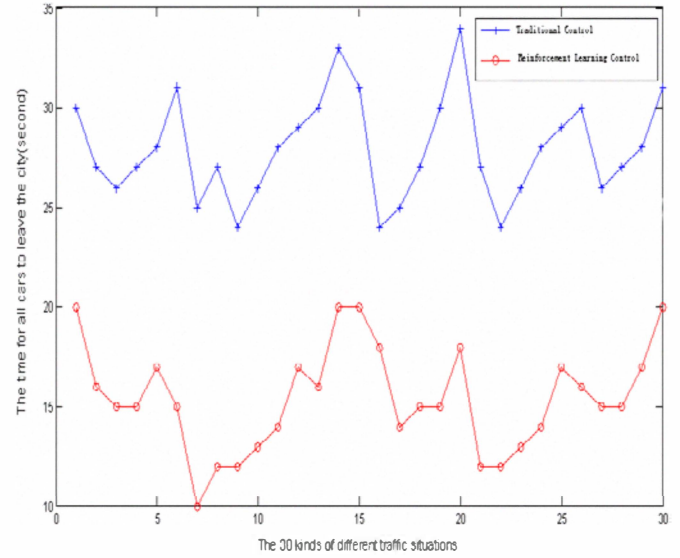


Figure4. The comparison between traditional control and multiagent reinforcement learning based control

V. CONCLUSIONS

In this work, we propose a multiagent reinforcement learning based traffic signal control method. We define the multiagent reinforcement learning model of the traffic signal control problem, and propose the algorithm for discovering the optimal policy. We also implement a simulation environment to demonstrate obviously the control effect of our method, and the experiment results show our methods can significantly reduce the time of all vehicles to leave the urban traffic network.

ACKNOWLEDGMENT

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